

HydroTech Engineering Pvt. Ltd.

CIN: U45400GJ2011PTC066214 | Registered Office: Surat, Gujraat | Established: 2011

**TECHNICAL & PRICE BID SUBMISSION for Tender Ref. GWSSB/WTP-SURAT/2026/DBO-009 Design, Build and Operate (DBO) of 100 MLD Water Treatment Plant with 10-Year O&M, at
Surat**

Field	Details
Bidder Reference	RESPONSE_DOC_S2
Company Name	HydroTech Engineering Pvt. Ltd.
Date of Submission	05-Mar-2026
Overall Status (Evaluation Committee)	TECHNICALLY QUALIFIED WITH MAJOR DEVIATIONS - Price Bid Opened; Deviations Under Committee Review

Table of Contents

1. Cover Letter & Company Profile
2. Eligibility Compliance Statement
3. Financial Standing
4. Detailed Project Experience (D&B and O&M) (incl. 4.1 Individual Project Write-ups)
5. Design Basis & Engineering Philosophy
6. Raw Water Pumping Station - Technical Specification
7. Clariflocculator - Technical Specification
8. Rapid Gravity Filters - Technical Specification
9. Chlorination System - Technical Specification
10. Sludge Handling - Technical Specification
11. Clear Water Reservoir & Pumping / Rising Main - Technical Specification
12. SCADA & Instrumentation - Technical Specification
13. Equipment OEM / Sub-Vendor Compliance
14. Quality Assurance Plan (QAP) - Inspection & Test Points
15. Health, Safety & Environment (HSE) Plan
16. Project Execution Methodology (D&B Phase)
17. Detailed Project Schedule (D&B Phase)
18. Manpower Deployment Plan (D&B Phase)
19. Plant, Machinery & Equipment Deployment
20. Mandatory Spares List
21. Testing & Commissioning Protocol
- 21A. Project Risk Register
- 21B. Terms & Conditions Compliance Matrix
- 21C. Key Personnel Profiles
- 21D. Sub-Vendor & OEM Profiles
22. O&M Methodology & Approach (10-Year)

23. Training & Warranty / Defect Liability
24. Deviations from Tender Requirements
25. Clarifications Sought & Bidder Responses
26. Priced Bill of Quantities (BOQ) - Stage 1 (Initial D&B Price)
27. Priced Bill of Quantities (BOQ) - Stage 2 (Final D&B Price)
28. O&M Price Schedule (10-Year, Year-wise) & Combined Cost Evaluation
29. Payment Terms Compliance Statement
30. List of Annexures
31. Compliance Undertaking
32. Evaluation Committee Remarks

1. Cover Letter & Company Profile

HydroTech Engineering Pvt. Ltd. confirms having read and fully understood the Instructions to Bidders, General and Special Conditions of Contract, Scope of Work, Technical Specifications, Eligibility Criteria, Evaluation Criteria, O&M Requirements, and all other Parts of Tender Document GWSSB/WTP-SURAT/2026/DBO-009, and hereby submits its offer for the Design, Build and Operate (DBO) of the 100 MLD Water Treatment Plant at Surat, on the terms set out in this bid.

HydroTech Engineering Pvt. Ltd., headquartered at Surat, Gujraat, was established in 2011 and has built a portfolio of water and wastewater treatment infrastructure projects across Gujraat and neighbouring states, encompassing both Design-Build execution and long-term Operation & Maintenance contracts. The company operates its own in-house process design cell, project management team, and O&M operations division, supplemented by long-term relationships with equipment OEMs for major process equipment supply.

The organizational structure for this project will be headed by a dedicated Project Manager for the D&B phase, transitioning to a dedicated O&M Manager from the commissioning phase onward, both reporting to the company's Head of Water Infrastructure Projects, with functional support from Process Design, Civil Engineering, Quality Assurance, HSE, Planning & Contracts, and Procurement departments.

2. Eligibility Compliance Statement

Criteria	Bidder Submission
EC-1: Average Annual Turnover (last 3 FY)	₹105.8 Cr / ₹112.2 Cr / ₹118.9 Cr Average: ₹112.30 Cr
EC-2: D&B Experience (50 MLD+ WTPs)	5 qualifying project(s) submitted - see Section 4 for details
EC-3: O&M Experience (3+ yr continuous, 50 MLD+)	1 qualifying reference(s) submitted - see Section 4 for details
EC-4: Net Worth	₹61.2 Cr (positive, per FY 2024-25 audited statement)
EC-5: Quality Certifications (ISO)	ISO 9001:2015, ISO 14001:2015, ISO 45001:2018 - all valid, all submitted
EC-8: EMD	₹2,00,00,000 Bank Guarantee, Bank of Baroda, valid 210 days - CONFORMS
Overall Eligibility Determination	COMPLIANT on EC-1, EC-2, EC-4 to EC-8; QUALIFIED WITH CONCERN on EC-3 - O&M reference contract is 3 years exactly (minimum threshold), no margin

3. Financial Standing

The financial statements submitted include audited Balance Sheets and Profit & Loss Accounts for FY 2022-23, 2023-24 and 2024-25, along with a Chartered Accountant's certificate confirming average annual turnover and a Banker's Solvency Certificate. A summary of key financial indicators is presented below.

Financial Parameter	FY 2022-23	FY 2023-24	FY 2024-25
Annual Turnover	₹105.8 Cr	₹112.2 Cr	₹118.9 Cr
Net Profit (after tax)	Refer audited P&L (Annexure-F)	Refer audited P&L (Annexure-F)	Refer audited P&L (Annexure-F)

Financial Parameter	FY 2022-23	FY 2023-24	FY 2024-25
Net Worth	Refer Annexure-F	Refer Annexure-F	₹61.2 Cr (positive, per FY 2024-25 audited statement)
Current Ratio	1.5	1.6	1.6
Debt-Equity Ratio	0.7	0.6	0.5

4. Detailed Project Experience (D&B and O&M)

The following table lists the bidder's relevant D&B and O&M project experience submitted in support of Eligibility Criteria EC-2 and EC-3 and the corresponding technical evaluation criteria. Projects marked with an asterisk (*) are below the relevant eligibility threshold and are included for information only.

Project Name	Location	Capacity/Type	Client	Year	Scope Summary
100 MLD WTP, Vadodara *	Vadodara, Gujraat	100 MLD, Conventional	VMSS	2022	Turnkey D&B of conventional WTP with clariflocculators and RGF filters.
60 MLD WTP, Rajkot	Rajkot, Gujraat	60 MLD, Conventional	RMC	2021	D&B of WTP with 8-year O&M contract, completed on schedule.
150 MLD WTP, Ahmedabad Extension *	Ahmedabad, Gujraat	150 MLD (extension), Conventional	AMC	2023	Extension of existing WTP with new clariflocculator train and upgraded filtration.
50 MLD WTP, Bhavnagar	Bhavnagar, Gujraat	50 MLD, Conventional	GWSSB	2020	D&B of grassroots WTP including raw water pumping and 8.5km rising main.
80 MLD WTP, Jamnagar	Jamnagar, Gujraat	80 MLD, Conventional	JMC	2024	Recently commissioned WTP, identical process scope to present tender.
30 MLD WTP, Junagadh *	Junagadh, Gujraat	30 MLD, Conventional	GWSSB	2019	D&B of WTP; below the EC-2 100 MLD single-project alternate threshold.
100 MLD WTP O&M Contract, Surat (Ongoing) *	Surat, Gujraat	100 MLD, O&M only	SMC	2019-ongoing	Ongoing 10-year O&M contract, 6 years completed at bid date, providing strong O&M track record.
120 MLD WTP, Nagpur *	Nagpur, Maharashtra	120 MLD, Conventional	NMC	2022	Interstate D&B project demonstrating capability outside home state.
40 MLD WTP, Gandhinagar	Gandhinagar, Gujraat	40 MLD, Conventional	GWSSB	2018	Older-generation D&B project with subsequent 5-year O&M contract, now concluded.
200 MLD WTP, Pune (JV Project) *	Pune, Maharashtra	200 MLD, Conventional	PMC	2023	Executed as JV lead partner; largest single project by capacity in bidder's portfolio.
25 MLD WTP, Anand *	Anand, Gujraat	25 MLD, Conventional	GWSSB	2017	Below the EC-2 50 MLD eligibility threshold for D&B experience.
Effluent Treatment Plant, Dahej Industrial Area *	Bharuch, Gujraat	40 MLD, ETP (not water treatment)	Private Industrial Client	2021	Related but not directly qualifying wastewater/effluent treatment experience.
70 MLD WTP O&M Contract, Vapi	Vapi, Gujraat	70 MLD, O&M only	GWSSB	2020-ongoing	Ongoing O&M contract, 5 years completed at bid date.
55 MLD WTP, Mehsana	Mehsana, Gujraat	55 MLD, Conventional	GWSSB	2021	D&B project with minor scope variation (membrane pre-

Project Name	Location	Capacity/Type	Client	Year	Scope Summary
					treatment pilot).
15 MLD WTP, Idar *	Sabarkantha, Gujraat	15 MLD, Conventional	GWSSB	2016	Small-capacity WTP, below EC-2 threshold, included for portfolio completeness.

Total qualifying D&B (50 MLD+) projects submitted by HydroTech Engineering Pvt. Ltd.: 5. Total qualifying continuous 3-year+ O&M (50 MLD+) references submitted: 1. Detailed completion certificates, O&M performance records, and client reference letters for each qualifying project are enclosed as Annexure-C.

4.1 Individual Project Reference Write-ups

100 MLD WTP, Vadodara (below eligibility threshold)

Client: VMSS | Location: Vadodara, Gujraat | Capacity/Type: 100 MLD, Conventional | Year: 2022

Turnkey D&B of conventional WTP with clariflocculators and RGF filters. HydroTech Engineering Pvt. Ltd. was engaged as the Contractor for this project. The project was executed as a smaller-scope or shorter-duration reference assignment, included here to demonstrate the bidder's broader water infrastructure execution capability. A completion certificate/performance record and client reference letter for this project are enclosed in Annexure-C, and referees are available for direct verification by GWSSB upon request.

60 MLD WTP, Rajkot

Client: RMC | Location: Rajkot, Gujraat | Capacity/Type: 60 MLD, Conventional | Year: 2021

D&B of WTP with 8-year O&M contract, completed on schedule. HydroTech Engineering Pvt. Ltd. was engaged as the Design-Build Contractor for this project. The project was executed in full compliance with the client's technical specifications and, where applicable, has consistently met contractual performance guarantees. A completion certificate/performance record and client reference letter for this project are enclosed in Annexure-C, and referees are available for direct verification by GWSSB upon request.

150 MLD WTP, Ahmedabad Extension (below eligibility threshold)

Client: AMC | Location: Ahmedabad, Gujraat | Capacity/Type: 150 MLD (extension), Conventional | Year: 2023

Extension of existing WTP with new clariflocculator train and upgraded filtration. HydroTech Engineering Pvt. Ltd. was engaged as the Contractor for this project. The project was executed as a smaller-scope or shorter-duration reference assignment, included here to demonstrate the bidder's broader water infrastructure execution capability. A completion certificate/performance record and client reference letter for this project are enclosed in Annexure-C, and referees are available for direct verification by GWSSB upon request.

50 MLD WTP, Bhavnagar

Client: GWSSB | Location: Bhavnagar, Gujraat | Capacity/Type: 50 MLD, Conventional | Year: 2020

D&B of grassroots WTP including raw water pumping and 8.5km rising main. HydroTech Engineering Pvt. Ltd. was engaged as the Design-Build Contractor for this project. The project was executed in full compliance with the client's technical specifications and, where applicable, has consistently met contractual performance guarantees. A completion certificate/performance record and client reference letter for this project are enclosed in Annexure-C, and referees are available for direct verification by GWSSB upon request.

80 MLD WTP, Jamnagar

Client: JMC | Location: Jamnagar, Gujraat | Capacity/Type: 80 MLD, Conventional | Year: 2024

Recently commissioned WTP, identical process scope to present tender. HydroTech Engineering Pvt. Ltd. was engaged as the Design-Build Contractor for this project. The project was executed in full compliance with the client's technical specifications and, where applicable, has consistently met contractual performance guarantees. A completion certificate/performance record and client reference letter for this project are enclosed in Annexure-C, and referees are available for direct verification by GWSSB upon request.

30 MLD WTP, Junagadh (below eligibility threshold)

Client: GWSSB | Location: Junagadh, Gujraat | Capacity/Type: 30 MLD, Conventional | Year: 2019

D&B of WTP; below the EC-2 100 MLD single-project alternate threshold. HydroTech Engineering Pvt. Ltd. was engaged as the Contractor for this project. The project was executed as a smaller-scope or shorter-duration reference assignment, included here to demonstrate the bidder's broader water infrastructure execution capability. A completion certificate/performance record and client reference letter for this project are enclosed in Annexure-C, and referees are available for direct verification by GWSSB upon request.

100 MLD WTP O&M Contract, Surat (Ongoing) (below eligibility threshold)

Client: SMC | Location: Surat, Gujraat | Capacity/Type: 100 MLD, O&M only | Year: 2019-ongoing

Ongoing 10-year O&M contract, 6 years completed at bid date, providing strong O&M track record. HydroTech Engineering Pvt. Ltd. was engaged as the Contractor for this project. The project was executed as a smaller-scope or shorter-duration reference assignment, included here to demonstrate the bidder's broader water infrastructure execution capability. A completion certificate/performance record and client reference letter for this project are enclosed in Annexure-C, and referees are available for direct verification by GWSSB upon request.

120 MLD WTP, Nagpur (below eligibility threshold)

Client: NMC | Location: Nagpur, Maharashtra | Capacity/Type: 120 MLD, Conventional | Year: 2022

Interstate D&B project demonstrating capability outside home state. HydroTech Engineering Pvt. Ltd. was engaged as the Contractor for this project. The project was executed as a smaller-scope or shorter-duration reference assignment, included here to demonstrate the bidder's broader water infrastructure execution capability. A completion certificate/performance record and client reference letter for this project are enclosed in Annexure-C, and referees are available for direct verification by GWSSB upon request.

40 MLD WTP, Gandhinagar

Client: GWSSB | Location: Gandhinagar, Gujraat | Capacity/Type: 40 MLD, Conventional | Year: 2018

Older-generation D&B project with subsequent 5-year O&M contract, now concluded. HydroTech Engineering Pvt. Ltd. was engaged as the Design-Build Contractor for this project. The project was executed in full compliance with the client's technical specifications and, where applicable, has consistently met contractual performance guarantees. A completion certificate/performance record and client reference letter for this project are enclosed in Annexure-C, and referees are available for direct verification by GWSSB upon request.

200 MLD WTP, Pune (JV Project) (below eligibility threshold)

Client: PMC | Location: Pune, Maharashtra | Capacity/Type: 200 MLD, Conventional | Year: 2023

Executed as JV lead partner; largest single project by capacity in bidder's portfolio. HydroTech Engineering Pvt. Ltd. was engaged as the Contractor for this project. The project was executed as a smaller-scope or shorter-duration reference assignment, included here to demonstrate the bidder's broader water infrastructure execution capability. A completion certificate/performance record and client reference letter for this project are enclosed in Annexure-C, and referees are available for direct verification by GWSSB upon request.

25 MLD WTP, Anand (below eligibility threshold)

Client: GWSSB | Location: Anand, Gujraat | Capacity/Type: 25 MLD, Conventional | Year: 2017

Below the EC-2 50 MLD eligibility threshold for D&B experience. HydroTech Engineering Pvt. Ltd. was engaged as the Contractor for this project. The project was executed as a smaller-scope or shorter-duration reference assignment, included here to demonstrate the bidder's broader water infrastructure execution capability. A completion certificate/performance record and client reference letter for this project are enclosed in Annexure-C, and referees are available for direct verification by GWSSB upon request.

Effluent Treatment Plant, Dahej Industrial Area (below eligibility threshold)

Client: Private Industrial Client | Location: Bharuch, Gujraat | Capacity/Type: 40 MLD, ETP (not water treatment) | Year: 2021

Related but not directly qualifying wastewater/effluent treatment experience. HydroTech Engineering Pvt. Ltd. was engaged as the Contractor for this project. The project was executed as a smaller-scope or shorter-duration reference assignment, included here to demonstrate the bidder's broader water infrastructure execution capability. A completion certificate/performance record and client reference letter for this project are enclosed in Annexure-C, and referees are available for direct verification by GWSSB upon request.

70 MLD WTP O&M Contract, Vapi

Client: GWSSB | Location: Vapi, Gujraat | Capacity/Type: 70 MLD, O&M only | Year: 2020-ongoing

Ongoing O&M contract, 5 years completed at bid date. HydroTech Engineering Pvt. Ltd. was engaged as the O&M Contractor for this project, with continuous operational responsibility as detailed in the O&M performance record enclosed in Annexure-C. The project was executed in full compliance with the client's technical specifications and, where applicable, has consistently met contractual performance guarantees. A completion certificate/performance record and client reference letter for this project are enclosed in Annexure-C, and referees are available for direct verification by GWSSB upon request.

55 MLD WTP, Mehsana

Client: GWSSB | Location: Mehsana, Gujraat | Capacity/Type: 55 MLD, Conventional | Year: 2021

D&B project with minor scope variation (membrane pre-treatment pilot). HydroTech Engineering Pvt. Ltd. was engaged as the Design-Build Contractor for this project. The project was executed in full compliance with the client's technical specifications and, where applicable, has consistently met contractual performance guarantees. A completion certificate/performance record and client reference letter for this project are enclosed in Annexure-C, and referees are available for direct verification by GWSSB upon request.

15 MLD WTP, Idar (below eligibility threshold)

Client: GWSSB | Location: Sabarkantha, Gujraat | Capacity/Type: 15 MLD, Conventional | Year: 2016

Small-capacity WTP, below EC-2 threshold, included for portfolio completeness. HydroTech Engineering Pvt. Ltd. was engaged as the Contractor for this project. The project was executed as a smaller-scope or shorter-duration reference assignment, included here to demonstrate the bidder's broader water infrastructure execution capability. A completion certificate/performance record and client reference letter for this project are enclosed in Annexure-C, and referees are available for direct verification by GWSSB upon request.

5. Design Basis & Engineering Philosophy

HydroTech Engineering Pvt. Ltd. has carried out preliminary process and engineering studies for the proposed 100 MLD Conventional Water Treatment Plant at Kosad, Surat, based on the design basis parameters, raw water characteristics data (Annexure L-2), and site data (Annexure L-1) provided in the Tender Document, supplemented by the bidder's own site visit during the pre-bid period. The design basis adopted is consistent with the CPHEEO Manual on Water Supply and Treatment, relevant Indian Standards, and GWSSB's Standard Specifications (Parts L-A and L-B of the BASE_DOC).

The raw water quality data reviewed indicates a wide seasonal variation in turbidity, from approximately 10 NTU in the lean season to over 1000 NTU during monsoon peak flows. The treatment train has accordingly been sized with adequate flexibility in coagulant dosing (jar-test-optimized) and clariflocculator surface loading to consistently achieve the guaranteed treated water turbidity of ≤ 1 NTU across this full seasonal range.

The overall plant hydraulic profile has been designed to operate substantially by gravity flow from the clariflocculator outlet through to the Clear Water Reservoir, minimizing intermediate pumping requirements and associated power consumption, consistent with the specific power consumption guarantee of ≤ 0.55 kWh per kL set out in Part H of the BASE_DOC.

Seismic considerations for the Surat region (Zone III per IS 1893) have been incorporated into the structural design philosophy for all major water-retaining structures, with crack-width control per IS 3370 to ensure long-term watertightness and durability over the specified 30-year civil design life.

The plant layout has been developed to provide adequate operational access for maintenance vehicles and equipment removal at all major process units, with the chlorination and chemical storage areas segregated per the safety requirements of Part L-A (CS-14) and located with due consideration to prevailing wind direction relative to the administrative building and site boundary.

6. Raw Water Pumping Station - Technical Specification

Parameter	Tender Requirement / Bidder's Offer
Number of Pumps	4 Working + 2 Standby (vertical turbine type)
Individual Pump Capacity	625 lps @ 30m head (each)
Pump Efficiency	≥ 78% at duty point
Motor Rating	IE3 premium efficiency, sized per pump duty
VFDs	Provided for flow modulation on 2 of 4 working pumps
Screening	Mechanically raked bar screens, 2 working + 1 standby
Pump House Type	RCC, semi-underground with dry/wet well configuration
Control Philosophy	Level-based auto start/stop with SCADA integration
Standby Power	Connected to DG backup per E.8
Noise Level	≤ 75 dB(A) at 1m from pump house wall

7. Clariflocculator - Technical Specification

Parameter	Tender Requirement / Bidder's Offer
Number of Units	4 (each rated for 25 MLD)
Type	Circular, mechanically agitated, with tube settler modules
Surface Loading Rate	$\leq 1.5 \text{ m}^3/\text{m}^2/\text{hr}$
Flocculation Detention Time	≥ 30 minutes
Clarification Detention Time	2.5 - 3 hours
Sludge Scraping Mechanism	Motorized central drive with scraper
Sludge Withdrawal	Automated, timer/turbidity-based
Weir Loading Rate	$\leq 300 \text{ m}^3/\text{m}/\text{day}$
Material - Mechanism	SS 304/316 for wetted parts
Material - Structure	RCC with crystalline waterproofing

8. Rapid Gravity Filters - Technical Specification

Proposed filter media configuration: Sand 55cm / Anthracite 25cm (BELOW specified minimum range of 60-90cm / 30-45cm).

Parameter	Tender Requirement / Bidder's Offer
Number of Filter Beds	16 Nos. (each approx. 45 sqm)
Filter Media	Dual media - sand + anthracite
Sand Depth	60-90 cm
Anthracite Depth	30-45 cm
Filtration Rate	5-6 m ³ /m ² /hr
Backwash Rate	36-44 m ³ /m ² /hr
Air Scour Rate	0.6-0.9 m ³ /m ² /min
Backwash System	Air scour + water wash, automatic sequential
Underdrain	Nozzle-type, HDPE/PP nozzles
Filter Control	PLC-based automatic control with SCADA integration

9. Chlorination System - Technical Specification

Proposed disinfection system: Sodium hypochlorite dosing offered in place of gas chlorination, without an explicit deviation declaration in the covering letter (identified by Evaluation Committee during technical scrutiny).

Parameter	Tender Requirement / Bidder's Offer
Primary Disinfectant	Chlorine gas (Cl ₂), automatic vacuum feed
Storage	Ton container, minimum 15 days consumption
Safety System	Gas scrubber, leak detectors, emergency kit per CPCB norms
Dosing Points	Post-filtration (primary), pre-chlorination (optional)
Contact Time	≥ 30 minutes at peak flow
Residual Monitoring	Online analyzers at CWR inlet and distribution points
Ventilation	Forced draft, discharge through scrubber

10. Sludge Handling - Technical Specification

Parameter	Tender Requirement / Bidder's Offer
Thickening	Gravity thickener, 2 Nos.
Dewatering	Centrifuge or Filter Press, min. 20% cake dryness
Drying Beds	Provided as backup for monsoon peak load
Disposal	As per GPCB norms, Contractor responsible during O&M
Sludge Production Estimate	1.5-2.5% of raw water volume (turbidity dependent)

11. Clear Water Reservoir & Pumping / Rising Main - Technical Specification

Parameter	Tender Requirement / Bidder's Offer
CWR Capacity	16.7 ML minimum (4 hours at design flow)
CWR Type	RCC underground/semi-underground water-retaining structure
Clear Water Pumps	5 Working + 2 Standby (horizontal split-case)
Individual Pump Capacity	500 lps @ 45m head (each)
Rising Main Length	Approximately 8.5 km
Rising Main Material	MS (welded) / Ductile Iron
Rising Main Diameter	1400mm (indicative)
Surge Protection	Surge vessels / relief valves per transient analysis

12. SCADA & Instrumentation - Technical Specification

Parameter	Tender Requirement / Bidder's Offer
Architecture	PLC-based distributed control with central SCADA server
Instrumentation	Flow, level, turbidity, chlorine, pH at all critical points
Communication	Fiber optic backbone between process areas and control room
Remote Monitoring	Integration with GWSSB central monitoring cell
Data Historian	Minimum 5-year retention
Redundancy	Dual redundant SCADA server, UPS-backed control systems

13. Equipment OEM / Sub-Vendor Compliance

Component	Proposed OEM / Make
Vertical Turbine Pumps (Raw Water)	Per Annexure L-5 Approved Makes List cross-reference (Annexure-E of this bid)
Horizontal Split Case Pumps (Clear Water)	Per Annexure L-5 Approved Makes List cross-reference (Annexure-E of this bid)
Clariflocculator Mechanism	Per Annexure L-5 Approved Makes List cross-reference (Annexure-E of this bid)
Rapid Gravity Filter System	Per Annexure L-5 Approved Makes List cross-reference (Annexure-E of this bid)
Chlorination System	Per Annexure L-5 Approved Makes List cross-reference (Annexure-E of this bid)
Sludge Dewatering Equipment	Per Annexure L-5 Approved Makes List cross-reference (Annexure-E of this bid)
MS/DI Pipes (Rising Main)	Per Annexure L-5 Approved Makes List cross-reference (Annexure-E of this bid)
HT Switchgear & Transformers	Per Annexure L-5 Approved Makes List cross-reference (Annexure-E of this bid)
DG Sets	Per Annexure L-5 Approved Makes List cross-reference (Annexure-E of this bid)
SCADA/PLC System	Per Annexure L-5 Approved Makes List cross-reference (Annexure-E of this bid)
Field Instrumentation (Flow/Level/Quality Analyzers)	Per Annexure L-5 Approved Makes List cross-reference (Annexure-E of this bid)
MCC Panels	Per Annexure L-5 Approved Makes List cross-reference (Annexure-E of this bid)
Cables	Per Annexure L-5 Approved Makes List cross-reference (Annexure-E of this bid)
Laboratory Equipment	Per Annexure L-5 Approved Makes List cross-reference (Annexure-E of this bid)
Valves (Gate/Butterfly/Non-return)	Per Annexure L-5 Approved Makes List cross-reference (Annexure-E of this bid)

14. Quality Assurance Plan (QAP) - Inspection & Test Points

Quality Assurance for this project will be governed by a project-specific Quality Assurance Plan (QAP), submitted for GWSSB approval within 30 days of the Notice to Proceed, covering incoming material inspection, in-process manufacturing inspection at OEM works, and site civil/mechanical/electrical erection and testing quality checkpoints, consistent with the hold/witness point framework referenced in Part L-A (Civil Specifications) of the BASE_DOC.

S.No.	Inspection / Test Point	Hold/Witness/Review Classification
1	Incoming inspection - cement, steel (mill TC verification)	Hold Point - GWSSB/TPIA witness mandatory
2	Incoming inspection - pipe material (MS/DI) mechanical properties	Witness Point - GWSSB/TPIA may witness on notice
3	Concrete mix design approval & trial mix results	Review Point - documentary review by QA Engineer
4	Reinforcement placement inspection prior to concrete pour	Hold Point - GWSSB/TPIA witness mandatory
5	Hydrostatic (water) test - all water-retaining structures	Witness Point - GWSSB/TPIA may witness on notice
6	Waterproofing application inspection (crystalline/membrane)	Review Point - documentary review by QA Engineer
7	Structural steel galvanization thickness (DFT) check	Hold Point - GWSSB/TPIA witness mandatory
8	Pump - factory performance test (IS 9137) witness	Witness Point - GWSSB/TPIA may witness on notice
9	Motor - routine test certificate review	Review Point - documentary review by QA Engineer
10	Clariflocculator mechanism - factory inspection	Hold Point - GWSSB/TPIA witness mandatory
11	Filter media - sieve analysis & effective size verification	Witness Point - GWSSB/TPIA may witness on notice
12	Filter underdrain - nozzle spacing & level check	Review Point - documentary review by QA Engineer
13	Chlorination system - factory acceptance test (FAT)	Hold Point - GWSSB/TPIA witness mandatory
14	Gas leak detection system - functional test	Witness Point - GWSSB/TPIA may witness on notice
15	Sludge dewatering equipment - factory performance test	Review Point - documentary review by QA Engineer
16	Rising main pipe - hydrostatic pressure test (section-wise)	Hold Point - GWSSB/TPIA witness mandatory
17	Rising main - joint quality inspection (welded/rubber ring)	Witness Point - GWSSB/TPIA may witness on notice
18	Electrical panel - factory inspection & routine test	Review Point - documentary review by QA Engineer
19	Cable - continuity & insulation resistance test	Hold Point - GWSSB/TPIA witness mandatory
20	Earthing system - resistance measurement	Witness Point - GWSSB/TPIA may witness on notice
21	SCADA/PLC - functional test (factory)	Review Point - documentary review by QA Engineer
22	SCADA/PLC - I/O loop check (site)	Hold Point - GWSSB/TPIA witness mandatory
23	Instrument calibration certificates (flow/level/quality analyzers)	Witness Point - GWSSB/TPIA may witness on notice

S.No.	Inspection / Test Point	Hold/Witness/Review Classification
24	Laboratory equipment - installation & calibration verification	Review Point - documentary review by QA Engineer
25	Civil works - concrete cube compressive strength test	Hold Point - GWSSB/TPIA witness mandatory
26	Civil works - soil compaction test (proctor density)	Witness Point - GWSSB/TPIA may witness on notice
27	Painting/coating - DFT verification (structural steel)	Review Point - documentary review by QA Engineer
28	Final inspection & pre-commissioning checklist sign-off	Hold Point - GWSSB/TPIA witness mandatory
29	Performance Guarantee Test Run - data verification	Witness Point - GWSSB/TPIA may witness on notice
30	As-built drawing accuracy verification against site condition	Review Point - documentary review by QA Engineer

15. Health, Safety & Environment (HSE) Plan

HydroTech Engineering Pvt. Ltd. maintains a documented Health, Safety & Environment Management System aligned with ISO 45001:2018 and ISO 14001:2015, applied to all water/wastewater infrastructure project sites, with particular attention to the specific hazards associated with chlorine gas handling. The HSE Plan for this project will be site-specific, reviewed and approved by GWSSB prior to mobilization, and enforced through a dedicated site HSE team. The following checklist summarizes the key HSE controls proposed for both the construction (D&B) and operation (O&M) phases of the project.

S.No.	HSE Control Measure	Responsibility
1	Site-specific HSE Plan approved before mobilization	Site HSE Officer
2	Daily Tool Box Talk (TBT) records maintained	Project Manager / O&M Manager
3	Permit-to-Work system for excavation, hot works, confined space entry	Site Engineer (all disciplines)
4	PPE compliance monitoring (helmets, safety harness, gas masks near chlorine area)	HSE Officer + Site Supervisors
5	Confined space entry procedure (CWR, clariflocculator, sludge tanks)	Site HSE Officer
6	Chlorine leak emergency response drill (quarterly)	Project Manager / O&M Manager
7	First-aid station and trained first-aiders on site	Site Engineer (all disciplines)
8	Fire extinguishers positioned at all high-risk zones	HSE Officer + Site Supervisors
9	Emergency response & evacuation plan displayed and rehearsed	Site HSE Officer
10	Monthly safety audit and non-conformance tracking	Project Manager / O&M Manager
11	Incident/near-miss reporting and root-cause analysis (RCA) system	Site Engineer (all disciplines)
12	Crane & lifting equipment - third-party certification	HSE Officer + Site Supervisors
13	Excavation safety - shoring/trench box for depth > 1.5m	Site HSE Officer
14	Electrical safety - LOTO procedure for live equipment isolation	Project Manager / O&M Manager
15	Environmental management - dust and noise control during construction	Site Engineer (all disciplines)
16	Chemical handling SOP (alum, chlorine, polyelectrolyte)	HSE Officer + Site Supervisors
17	Worker health check-up (pre-employment and periodic)	Site HSE Officer
18	Safety induction training for all site personnel	Project Manager / O&M Manager
19	Zero Lost-Time Injury (LTI) target with monthly HSE KPI reporting	Site Engineer (all disciplines)
20	Waste management - segregation and disposal per GPCB norms	HSE Officer + Site Supervisors

16. Project Execution Methodology (D&B Phase)

HydroTech Engineering Pvt. Ltd. proposes to execute this project using a parallel-path methodology wherein civil works, long-lead equipment procurement (pumps, clariflocculator mechanisms, filter systems, chlorination equipment), rising main procurement, and detailed engineering proceed concurrently from Day 1 of mobilization, in order to achieve the approximately 20-month Design-Build schedule while managing interface risk through a dedicated Planning Engineer using a detailed Level-3 project schedule. Monthly progress review meetings will be held with GWSSB, supported by weekly internal progress tracking.

Key execution phases comprise: (i) Mobilization & Engineering, (ii) Civil Works, (iii) Equipment Procurement & Manufacturing, (iv) Rising Main Laying, (v) Erection & Installation, (vi) Testing, Pre-Commissioning & Commissioning, (vii) 30-Day Performance Guarantee Test Run, and (viii) Provisional Acceptance & Handover to O&M. Interfaces between civil and mechanical/electrical erection teams will be managed through a joint interface schedule reviewed weekly.

17. Detailed Project Schedule (D&B Phase)

The overall D&B duration proposed is 552 days from Notice to Proceed (NTP) to Provisional Acceptance, against the Tender's indicative schedule of approximately 600 days (Part J of BASE_DOC). The detailed activity-wise schedule (Level-2) is presented below.

Activity	Planned Completion (Days from NTP)
Notice to Proceed (NTP) issuance	0
Contract signing & kick-off meeting	6
Mobilization of site team & establishment	18
Topographic & geotechnical survey	28
Preparation of Design Basis Report (DBR)	41
Process Flow Diagram (PFD) submission	51
Hydraulic Profile submission	55
GA drawings - all major structures	83
Structural design & drawing approval cycle	101
Client review & drawing approval - complete	110
Site clearance & grading	55
Raw Water Pumping Station - foundation	138
Raw Water Pumping Station - civil complete	221
WTP - clariflocculator civil structures	258
WTP - filter civil structures	276
WTP - CWR civil structures	294
WTP - chlorination building civil	239
WTP - sludge handling civil	258
Administrative & laboratory building - civil	322
Boundary wall & internal roads	350
Placing of pump orders (OEM)	41
Pump manufacturing & FAT	193
Pump delivery to site	212
Placing of clariflocculator mechanism order	55
Clariflocculator mechanism - manufacturing & delivery	239
Placing of filter system order	55
Filter media & underdrain - delivery	258
Placing of chlorination system order	55
Chlorination system - delivery & installation	294
Placing of sludge dewatering equipment order	69
Sludge dewatering equipment - delivery	276
Rising main - pipe procurement	110
Rising main - trenching & laying (Section 1)	276
Rising main - trenching & laying (Section 2)	331
Rising main - trenching & laying (Section 3)	386
Rising main - hydrostatic testing	414
Rising main - backfilling & road restoration	432
Raw Water Pumping Station - equipment erection	294
WTP mechanical equipment erection	423
WTP electrical & MCC installation	442
SCADA/PLC installation & configuration	460
Clear Water Pumping Station - erection	423
Laboratory - equipment installation & commissioning	451
Pre-commissioning checks - all sub-systems	488
Individual equipment commissioning (dry/wet run)	506
Integrated commissioning - full process train	520
30-Day Performance Guarantee Test Run	552

Activity	Planned Completion (Days from NTP)
Punch-list closure & as-built documentation	543
Handover & training of GWSSB O&M staff	547
Provisional Acceptance & O&M commencement	552

18. Manpower Deployment Plan (D&B Phase)

Role / Designation	Qualification / Experience	Peak Headcount
Project Manager (D&B Phase)	B.E. Civil/Environmental, 15+ yrs WTP EPC experience	1
Deputy Project Manager	B.E. Civil/Mechanical, 10+ yrs	1
Design Manager - Process	B.E./M.Tech Environmental Engineering, process design specialist	1
Design Manager - Civil/Structural	B.E. Civil, structural design experience	1
Design Manager - Electrical/SCADA	B.E. Electrical, SCADA/instrumentation experience	1
Site Engineer - Civil	B.E. Civil, 8+ yrs	3
Site Engineer - Mechanical	B.E. Mechanical, 6+ yrs	2
Site Engineer - Electrical & Instrumentation	B.E. Electrical, 6+ yrs	2
QA/QC Engineer	B.E. with NDT/QC certification	2
HSE Officer	Diploma with NEBOSH/IOSH certification	2
Planning Engineer	B.E. with Primavera/MS Project expertise	1
Draughtsman (AutoCAD)	Diploma, AutoCAD proficient	2
Store & Logistics In-charge	Graduate, ERP/inventory management	1
Foreman - Civil	ITI/Diploma, 10+ yrs	3
Foreman - Mechanical/Electrical	ITI/Diploma, 10+ yrs	3
Skilled Fitters & Erectors	ITI certified	12
Skilled Welders (pipeline)	ITI Welder, certified (for rising main)	8
Skilled Electricians	ITI Electrician, licensed	8
Riggers & Crane Operators	Certified riggers/operators	5
Civil Masons & Carpenters	Skilled trade	15
Unskilled Labour	Local workforce	25
Testing & Commissioning Engineer	B.E., process/electrical commissioning specialist	2
SCADA/Instrumentation Commissioning Engineer	B.E./B.Tech, PLC/SCADA experience	2
Surveyor	Diploma, total station/GPS survey	2
Store Assistant	Graduate	2
Security Supervisor	Ex-serviceman preferred	2
Security Guards	Trained security personnel	7
Document Controller	Graduate, document management	1
Admin & HR Coordinator	Graduate	1
Environment Engineer	B.E./M.Sc. Environmental Science	1

19. Plant, Machinery & Equipment Deployment

S.No.	Construction Plant / Testing Equipment	Ownership / Source
1	Mobile Crane 50T capacity	Company owned
2	Mobile Crane 25T capacity	Hired, on long-term rental from empanelled vendor
3	Hydra Crane 14T capacity	Company owned
4	Excavator (JCB) 0.9 cum	Hired, on long-term rental from empanelled vendor
5	Pipe-laying excavator (long-reach)	Company owned
6	Concrete Batching Plant (30 cum/hr)	Hired, on long-term rental from empanelled vendor
7	Concrete Transit Mixer (6 cum)	Company owned
8	Vibrator - Needle type	Hired, on long-term rental from empanelled vendor
9	DG Set 250 kVA (construction power)	Company owned
10	DG Set 125 kVA (backup)	Hired, on long-term rental from empanelled vendor
11	Pipe bending/welding machine set (rising main)	Company owned
12	Trenching machine	Hired, on long-term rental from empanelled vendor
13	Dewatering pump set (construction phase)	Company owned
14	Hydrostatic pressure test pump/kit	Hired, on long-term rental from empanelled vendor
15	Insulation Resistance (IR) Tester	Company owned
16	Earth Resistance Tester	Hired, on long-term rental from empanelled vendor
17	Vibration & Noise Level Meter	Company owned
18	Water Quality Field Test Kit	Hired, on long-term rental from empanelled vendor
19	Total Station (survey)	Company owned
20	GPS Survey Equipment	Hired, on long-term rental from empanelled vendor

20. Mandatory Spares List

The following mandatory spares are offered as part of the D&B Contract Price, for GWSSB's use during the Defect Liability Period and the early years of the O&M period; priced separately in the BOQ (Section 26/27, Item 58).

S.No.	Spare Item	Quantity Offered
1	Raw water pump - spare impeller set	1
2	Raw water pump - spare mechanical seal kit	2
3	Clear water pump - spare impeller set	3
4	Clear water pump - spare mechanical seal kit	1
5	Motor - spare bearing set (all pump motors)	2
6	VFD - spare control card	3
7	Clarifloculator - spare scraper mechanism drive unit	1
8	Filter - spare nozzles (10% of installed)	2
9	Chlorinator - spare vacuum regulator	3
10	Chlorinator - spare injector assembly	1
11	Gas leak detector - spare sensor cartridge	2
12	Sludge dewatering - spare filter cloth / centrifuge bowl liner	3
13	MCC - spare contactor & relay set	1
14	SCADA - spare I/O card set	2
15	Instrumentation - spare turbidity/chlorine analyzer probes	3
16	Battery bank - 10% spare capacity	1
17	Rising main - spare pipe sections & fittings (2% of length)	2
18	Valve - spare actuator set	3
19	Cable - 5% spare length (power & control)	1
20	Laboratory - spare reagent/consumable starter kit	2

21. Testing & Commissioning Protocol

S.No.	Test / Check	Stage
1	Hydrostatic test - Clear Water Reservoir	Factory (at OEM works, pre-dispatch)
2	Hydrostatic test - Clariflocculator units	Factory (at OEM works, pre-dispatch)
3	Pump - performance curve verification (flow vs head vs efficiency)	Factory (at OEM works, pre-dispatch)
4	Pump - vibration & noise level measurement	Factory (at OEM works, pre-dispatch)
5	Motor - insulation resistance & winding resistance test	Factory (at OEM works, pre-dispatch)
6	Clariflocculator - mechanism no-load run test	Factory (at OEM works, pre-dispatch)
7	Clariflocculator - loaded run with jar-test-optimized dosing	Site - Pre-commissioning
8	Filter - individual bed backwash sequence test	Site - Pre-commissioning
9	Filter - filtered water turbidity verification (per bed)	Site - Pre-commissioning
10	Chlorination system - dosing accuracy verification	Site - Pre-commissioning
11	Chlorination system - gas leak detector functional test	Site - Pre-commissioning
12	Sludge dewatering - cake dryness verification test	Site - Pre-commissioning
13	Rising main - hydrostatic pressure test (full length, section-wise)	Site - Pre-commissioning
14	Rising main - surge/transient analysis field verification	Site - Pre-commissioning
15	Clear water pumps - performance curve verification	Site - Pre-commissioning
16	SCADA - point-to-point signal verification	Site - Pre-commissioning
17	SCADA - alarm/interlock functional test	Site - Pre-commissioning
18	DG backup - changeover time test	Site - Pre-commissioning
19	Earthing system - resistance measurement	Site - Pre-commissioning
20	Laboratory - calibration verification of all instruments	Site - Pre-commissioning
21	Water quality - baseline raw water testing (Day 0)	Site - Commissioning / PGT Run
22	Water quality - daily composite sampling during PGT run	Site - Commissioning / PGT Run
23	Specific power consumption - daily monitoring during PGT run	Site - Commissioning / PGT Run
24	Specific chemical consumption - daily monitoring during PGT run	Site - Commissioning / PGT Run
25	Final joint inspection & Provisional Acceptance sign-off	Site - Commissioning / PGT Run

21A. Project Risk Register

The following risk register identifies the key risks anticipated for this project, their category, and the mitigation measures proposed by HydroTech Engineering Pvt. Ltd., consistent with the Risk Allocation Matrix set out in Part H-A of the BASE_DOC.

Risk Description	Category	Mitigation Measure
Delay in pump/clariflocculator OEM manufacturing	Schedule	Early order placement within 45 days of NTP; monthly OEM progress review
Monsoon impact on civil works and rising main trenching	Schedule	Front-load foundation/civil work before monsoon; dewatering provisions for below-ground works
Raw water quality significantly worse than historical data	Technical	Design contingency in coagulant dosing capacity; jar-test-based optimization protocol during commissioning
Right-of-way delays for rising main across public roads	External	Early liaison with municipal/highway authorities; phased alignment approval
Groundwater ingress during excavation for CWR/clariflocculators	Technical	Dewatering system design with contingency capacity; phased excavation sequencing
Chlorine gas safety incident during construction/commissioning	Safety	Comprehensive HSE Plan (Section on HSE); quarterly leak drills; certified safety personnel
Price escalation in steel/cement/pipe material	Commercial	Firm-price D&B Contract accepted; internal hedging through forward material booking
Interface delay between civil and mechanical/electrical erection	Coordination	Weekly joint interface meetings; dedicated interface coordinator
Grid power connection delay for O&M phase	External	Early Discom liaison per Annexure L-1; DG backup available for interim commissioning power
O&M performance guarantee shortfall in early years	Performance	Extended commissioning/tuning period requested; process optimization support from OEMs
Sludge disposal site unavailability during O&M	Operational	Multiple disposal site options identified; sludge drying beds as interim buffer
Key O&M personnel attrition	Resource	Backup personnel identified; competitive retention package; annual training program
Design changes requested by Employer during execution	Scope	Formal change control process with defined cost/schedule impact assessment
Quality non-conformance at OEM works	Quality	TPIA witness at all critical hold points; NCR closure tracking system
Cyber-security vulnerability in SCADA system	Technical	Network segmentation; access control; periodic vulnerability assessment during O&M
Land subsidence/soil bearing capacity deviation from assumption	Technical	Detailed geotechnical investigation in first 30 days; design contingency in foundation sizing
Third-party utility damage during pipeline trenching	External	Utility mapping survey before excavation; permit-to-work for all digging
Testing/calibration equipment lapse	Quality	Calibration tracking register; NABL-calibrated instruments only
Documentation/as-built delay affecting handover	Administrative	Dedicated document controller; concurrent as-built markup during execution
WPI-linked O&M escalation exceeding budget assumptions	Commercial	Escalation capped at 7% p.a. per SCC-9; internal cost monitoring against this cap

21A.1 Contingency Planning

In addition to the risk-specific mitigation measures above, HydroTech Engineering Pvt. Ltd. maintains a general project contingency of 5% of the D&B Contract Value (internally budgeted, not separately billed to GWSSB) to absorb minor cost/schedule impacts from residual risks not individually itemized

above, and a dedicated Contingency Management Protocol under which any risk event with a potential impact exceeding ₹50 lakh or 15 days of schedule delay is escalated to the company's senior management within 48 hours of identification, with a mitigation/recovery plan submitted to GWSSB within 7 days.

21B. Terms & Conditions Compliance Matrix

The table below cross-references each major term of the BASE_DOC (Parts B, C and J) against the bidder's compliance status.

Clause	Requirement	Bidder Compliance
Bid Validity	180 days from Price Bid opening	Complied
D&B Completion Period	Approx. 20 months (600 days) from NTP	Complied - see Section 12 for bidder's proposed schedule
Liquidated Damages (D&B)	0.5% of D&B Contract Value per week, capped at 10%	Complied
Defect Liability Period (Civil / M&E)	24 months / 18 months from Provisional Acceptance	Complied
Performance Security	5% of D&B Price + 5% of NPV O&M Price, separate instruments	Complied
O&M Period	10 years from Provisional Acceptance	Complied
O&M Price Escalation Cap	WPI-linked, max 7% p.a. from Year 2	Complied
Power Cost during O&M	Borne by Employer	Complied - noted, Contractor responsible for efficient usage only
Chemicals & Consumables during O&M	Contractor responsibility, per specified consumption norms	Complied
Sludge Disposal during O&M	Contractor responsibility, GPCB-compliant	Complied
Performance Guarantee Test	30-day joint test run prior to Provisional Acceptance	Complied
Handback Condition (End of O&M)	Min. 5-year residual life for major M&E equipment	Complied
Insurance	CAR + TPL for D&B; appropriate O&M-phase insurance	Complied
Dispute Resolution	Arbitration under Arbitration & Conciliation Act, 1996	Complied
Force Majeure	As per GCC-28	Complied
Sub-contracting	Only with Employer's prior written approval	Complied - specialist O&M lab testing may be sub-contracted with approval
Integrity Pact	To be executed at Contract signing (Form K-4)	Complied - undertaking provided
Statutory Compliance (Labour)	EPF/ESI compliance for all site labour	Complied
Record Retention	Project records retained for 5 years post-O&M	Complied
Termination for Convenience	Employer's right to terminate with notice, per GCC-27	Noted and accepted

21C. Key Personnel Profiles

Brief profiles of the key personnel proposed to be deployed on this project (D&B and O&M phases) are provided below. Detailed CVs are enclosed as part of the technical proposal.

Project Manager (D&B Phase)

22+ years in water/wastewater treatment plant EPC; has led design-build of 6 WTPs of 50 MLD and above capacity.

This resource will be dedicated to the GWSSB Surat project for its full relevant duration and will not be shared concurrently with any other ongoing project of HydroTech Engineering Pvt. Ltd. above 50% allocation, except during ramp-down/ramp-up phases as mutually agreed with GWSSB's Engineer-in-Charge.

O&M Manager (Designate)

16 years in WTP operations; managed an operating 80 MLD plant for the preceding 6 years with consistent IS 10500 compliance.

This resource will be dedicated to the GWSSB Surat project for its full relevant duration and will not be shared concurrently with any other ongoing project of HydroTech Engineering Pvt. Ltd. above 50% allocation, except during ramp-down/ramp-up phases as mutually agreed with GWSSB's Engineer-in-Charge.

Design Head - Process

18 years in water treatment process design; M.Tech Environmental Engineering, specialist in clariflocculator and filtration system design.

This resource will be dedicated to the GWSSB Surat project for its full relevant duration and will not be shared concurrently with any other ongoing project of HydroTech Engineering Pvt. Ltd. above 50% allocation, except during ramp-down/ramp-up phases as mutually agreed with GWSSB's Engineer-in-Charge.

Design Head - Civil/Structural

15 years in water-retaining structure design; specialist in IS 3370 compliant RCC design.

This resource will be dedicated to the GWSSB Surat project for its full relevant duration and will not be shared concurrently with any other ongoing project of HydroTech Engineering Pvt. Ltd. above 50% allocation, except during ramp-down/ramp-up phases as mutually agreed with GWSSB's Engineer-in-Charge.

QA/QC Manager

13 years in water infrastructure quality assurance; certified in concrete and welding NDT methods.

This resource will be dedicated to the GWSSB Surat project for its full relevant duration and will not be shared concurrently with any other ongoing project of HydroTech Engineering Pvt. Ltd. above 50%

allocation, except during ramp-down/ramp-up phases as mutually agreed with GWSSB's Engineer-in-Charge.

HSE Manager

12 years in construction & process-plant HSE; NEBOSH IGC certified; specific chlorine safety training.

This resource will be dedicated to the GWSSB Surat project for its full relevant duration and will not be shared concurrently with any other ongoing project of HydroTech Engineering Pvt. Ltd. above 50% allocation, except during ramp-down/ramp-up phases as mutually agreed with GWSSB's Engineer-in-Charge.

Planning & Contracts Manager

14 years in EPC project planning; Primavera P6 certified; DBO contract administration experience.

This resource will be dedicated to the GWSSB Surat project for its full relevant duration and will not be shared concurrently with any other ongoing project of HydroTech Engineering Pvt. Ltd. above 50% allocation, except during ramp-down/ramp-up phases as mutually agreed with GWSSB's Engineer-in-Charge.

Commissioning Engineer-in-Charge

19 years in WTP testing & commissioning; has commissioned 8+ treatment plants including SCADA integration.

This resource will be dedicated to the GWSSB Surat project for its full relevant duration and will not be shared concurrently with any other ongoing project of HydroTech Engineering Pvt. Ltd. above 50% allocation, except during ramp-down/ramp-up phases as mutually agreed with GWSSB's Engineer-in-Charge.

21D. Sub-Vendor & OEM Profiles

HydroTech Engineering Pvt. Ltd. maintains long-standing relationships with the equipment OEMs proposed for this project. A brief profile of each major sub-vendor/OEM proposed is provided below.

Vertical Turbine Pumps (Raw Water)

This component will be sourced through a formal purchase order placed within 45 days of NTP, with design/drawing approval obtained from GWSSB prior to manufacture. Manufacturing will be subject to the inspection and test plan detailed in Section 14 (QAP), with hold points witnessed by GWSSB or its nominated Third-Party Inspection Agency as applicable.

Delivery lead time for this component is estimated at 5-8 months from date of order placement, factored into the detailed project schedule in Section 17. Payment to the sub-vendor/OEM will be released back-to-back with GWSSB's milestone payments to HydroTech Engineering Pvt. Ltd..

Horizontal Split Case Pumps (Clear Water)

This component will be sourced through a formal purchase order placed within 45 days of NTP, with design/drawing approval obtained from GWSSB prior to manufacture. Manufacturing will be subject to the inspection and test plan detailed in Section 14 (QAP), with hold points witnessed by GWSSB or its nominated Third-Party Inspection Agency as applicable.

Delivery lead time for this component is estimated at 5-8 months from date of order placement, factored into the detailed project schedule in Section 17. Payment to the sub-vendor/OEM will be released back-to-back with GWSSB's milestone payments to HydroTech Engineering Pvt. Ltd..

Clariflocculator Mechanism

This component will be sourced through a formal purchase order placed within 45 days of NTP, with design/drawing approval obtained from GWSSB prior to manufacture. Manufacturing will be subject to the inspection and test plan detailed in Section 14 (QAP), with hold points witnessed by GWSSB or its nominated Third-Party Inspection Agency as applicable.

Delivery lead time for this component is estimated at 5-8 months from date of order placement, factored into the detailed project schedule in Section 17. Payment to the sub-vendor/OEM will be released back-to-back with GWSSB's milestone payments to HydroTech Engineering Pvt. Ltd..

Rapid Gravity Filter System

This component will be sourced through a formal purchase order placed within 45 days of NTP, with design/drawing approval obtained from GWSSB prior to manufacture. Manufacturing will be subject to the inspection and test plan detailed in Section 14 (QAP), with hold points witnessed by GWSSB or its nominated Third-Party Inspection Agency as applicable.

Delivery lead time for this component is estimated at 5-8 months from date of order placement, factored into the detailed project schedule in Section 17. Payment to the sub-vendor/OEM will be released back-to-back with GWSSB's milestone payments to HydroTech Engineering Pvt. Ltd..

Chlorination System

This component will be sourced through a formal purchase order placed within 45 days of NTP, with design/drawing approval obtained from GWSSB prior to manufacture. Manufacturing will be subject to the inspection and test plan detailed in Section 14 (QAP), with hold points witnessed by GWSSB or its nominated Third-Party Inspection Agency as applicable.

Delivery lead time for this component is estimated at 5-8 months from date of order placement, factored into the detailed project schedule in Section 17. Payment to the sub-vendor/OEM will be released back-to-back with GWSSB's milestone payments to HydroTech Engineering Pvt. Ltd..

Sludge Dewatering Equipment

This component will be sourced through a formal purchase order placed within 45 days of NTP, with design/drawing approval obtained from GWSSB prior to manufacture. Manufacturing will be subject to the inspection and test plan detailed in Section 14 (QAP), with hold points witnessed by GWSSB or its nominated Third-Party Inspection Agency as applicable.

Delivery lead time for this component is estimated at 5-8 months from date of order placement, factored into the detailed project schedule in Section 17. Payment to the sub-vendor/OEM will be released back-to-back with GWSSB's milestone payments to HydroTech Engineering Pvt. Ltd..

MS/DI Pipes (Rising Main)

This component will be sourced through a formal purchase order placed within 45 days of NTP, with design/drawing approval obtained from GWSSB prior to manufacture. Manufacturing will be subject to the inspection and test plan detailed in Section 14 (QAP), with hold points witnessed by GWSSB or its nominated Third-Party Inspection Agency as applicable.

Delivery lead time for this component is estimated at 5-8 months from date of order placement, factored into the detailed project schedule in Section 17. Payment to the sub-vendor/OEM will be released back-to-back with GWSSB's milestone payments to HydroTech Engineering Pvt. Ltd..

HT Switchgear & Transformers

This component will be sourced through a formal purchase order placed within 45 days of NTP, with design/drawing approval obtained from GWSSB prior to manufacture. Manufacturing will be subject to the inspection and test plan detailed in Section 14 (QAP), with hold points witnessed by GWSSB or its nominated Third-Party Inspection Agency as applicable.

Delivery lead time for this component is estimated at 5-8 months from date of order placement, factored into the detailed project schedule in Section 17. Payment to the sub-vendor/OEM will be released back-to-back with GWSSB's milestone payments to HydroTech Engineering Pvt. Ltd..

DG Sets

This component will be sourced through a formal purchase order placed within 45 days of NTP, with design/drawing approval obtained from GWSSB prior to manufacture. Manufacturing will be subject to

the inspection and test plan detailed in Section 14 (QAP), with hold points witnessed by GWSSB or its nominated Third-Party Inspection Agency as applicable.

Delivery lead time for this component is estimated at 5-8 months from date of order placement, factored into the detailed project schedule in Section 17. Payment to the sub-vendor/OEM will be released back-to-back with GWSSB's milestone payments to HydroTech Engineering Pvt. Ltd..

SCADA/PLC System

This component will be sourced through a formal purchase order placed within 45 days of NTP, with design/drawing approval obtained from GWSSB prior to manufacture. Manufacturing will be subject to the inspection and test plan detailed in Section 14 (QAP), with hold points witnessed by GWSSB or its nominated Third-Party Inspection Agency as applicable.

Delivery lead time for this component is estimated at 5-8 months from date of order placement, factored into the detailed project schedule in Section 17. Payment to the sub-vendor/OEM will be released back-to-back with GWSSB's milestone payments to HydroTech Engineering Pvt. Ltd..

Field Instrumentation (Flow/Level/Quality Analyzers)

This component will be sourced through a formal purchase order placed within 45 days of NTP, with design/drawing approval obtained from GWSSB prior to manufacture. Manufacturing will be subject to the inspection and test plan detailed in Section 14 (QAP), with hold points witnessed by GWSSB or its nominated Third-Party Inspection Agency as applicable.

Delivery lead time for this component is estimated at 5-8 months from date of order placement, factored into the detailed project schedule in Section 17. Payment to the sub-vendor/OEM will be released back-to-back with GWSSB's milestone payments to HydroTech Engineering Pvt. Ltd..

MCC Panels

This component will be sourced through a formal purchase order placed within 45 days of NTP, with design/drawing approval obtained from GWSSB prior to manufacture. Manufacturing will be subject to the inspection and test plan detailed in Section 14 (QAP), with hold points witnessed by GWSSB or its nominated Third-Party Inspection Agency as applicable.

Delivery lead time for this component is estimated at 5-8 months from date of order placement, factored into the detailed project schedule in Section 17. Payment to the sub-vendor/OEM will be released back-to-back with GWSSB's milestone payments to HydroTech Engineering Pvt. Ltd..

Cables

This component will be sourced through a formal purchase order placed within 45 days of NTP, with design/drawing approval obtained from GWSSB prior to manufacture. Manufacturing will be subject to the inspection and test plan detailed in Section 14 (QAP), with hold points witnessed by GWSSB or its nominated Third-Party Inspection Agency as applicable.

Delivery lead time for this component is estimated at 5-8 months from date of order placement, factored into the detailed project schedule in Section 17. Payment to the sub-vendor/OEM will be released back-to-back with GWSSB's milestone payments to HydroTech Engineering Pvt. Ltd..

Laboratory Equipment

This component will be sourced through a formal purchase order placed within 45 days of NTP, with design/drawing approval obtained from GWSSB prior to manufacture. Manufacturing will be subject to the inspection and test plan detailed in Section 14 (QAP), with hold points witnessed by GWSSB or its nominated Third-Party Inspection Agency as applicable.

Delivery lead time for this component is estimated at 5-8 months from date of order placement, factored into the detailed project schedule in Section 17. Payment to the sub-vendor/OEM will be released back-to-back with GWSSB's milestone payments to HydroTech Engineering Pvt. Ltd..

Valves (Gate/Butterfly/Non-return)

This component will be sourced through a formal purchase order placed within 45 days of NTP, with design/drawing approval obtained from GWSSB prior to manufacture. Manufacturing will be subject to the inspection and test plan detailed in Section 14 (QAP), with hold points witnessed by GWSSB or its nominated Third-Party Inspection Agency as applicable.

Delivery lead time for this component is estimated at 5-8 months from date of order placement, factored into the detailed project schedule in Section 17. Payment to the sub-vendor/OEM will be released back-to-back with GWSSB's milestone payments to HydroTech Engineering Pvt. Ltd..

22. O&M Methodology & Approach (10-Year)

HydroTech Engineering Pvt. Ltd. proposes to commence O&M mobilization 60 days prior to the scheduled Provisional Acceptance date, to allow key O&M personnel to participate in the commissioning and Performance Guarantee Test Run activities and build direct familiarity with the specific plant configuration before assuming full operational responsibility.

Throughout the 10-year O&M period, HydroTech Engineering Pvt. Ltd. commits to the manpower levels specified in Part H.1 of the BASE_DOC as a minimum, with the detailed staffing plan set out in Section 13 of this proposal, organized on a 4-shift, 24x7x365 basis with defined escalation and emergency response protocols consistent with Part H.2 of the BASE_DOC.

A preventive maintenance schedule for all major mechanical and electrical equipment will be developed and submitted for GWSSB's review within 60 days of Provisional Acceptance, calibrated to the equipment OEMs' recommended maintenance intervals, and tracked through a computerized maintenance management system (CMMS) accessible to GWSSB for audit purposes.

22.1 O&M Performance Guarantee Compliance Statement

HydroTech Engineering Pvt. Ltd. confirms its intent to meet or exceed each of the performance guarantees set out in Part H of the BASE_DOC throughout the 10-year O&M period. The table below restates the guaranteed parameters alongside the bidder's own internal target (where more stringent than the minimum guarantee), for the Employer's reference.

Performance Parameter	GWSSB Guaranteed Value (Minimum)	Bidder's Internal Target
Plant Availability	≥ 98% annual average	≥ 98% annual average
Treated Water Turbidity	≤ 1 NTU, 95th percentile	≤ 1 NTU, 95th percentile
Residual Chlorine	0.2 - 0.5 mg/l	0.3 - 0.4 mg/l (tighter internal band)
IS 10500:2012 Compliance	100%	100%, with monthly internal audit in addition to third-party testing
Specific Power Consumption	≤ 0.55 kWh/kL	≤ 0.55 kWh/kL
Unplanned Shutdown	≤ 24 hours/annum	≤ 24 hours/annum

22.2 Illustrative O&M Reporting Format

Consistent with Part H.2 of the BASE_DOC, the following report types and contents are proposed to keep GWSSB fully informed of plant performance throughout the O&M period:

- Daily Operation Report: raw/treated water quality snapshot, chemical dosing rates, power consumption, shift log summary
- Weekly Maintenance Report: preventive maintenance activities completed, spares consumed, pending work orders
- Monthly O&M Performance Report: full performance guarantee parameter compliance summary, downtime log with root-cause analysis, chemical and power consumption trend
- Quarterly HSE Report: incident/near-miss summary, training conducted, drill records (including chlorine leak drill)
- Annual O&M Review Report: year-on-year performance trend, equipment condition assessment, capital spares replacement forecast for the following year
- Ad-hoc Incident Report: within 24 hours of any unplanned shutdown, water quality excursion, or safety incident

23. Training & Warranty / Defect Liability

HydroTech Engineering Pvt. Ltd. will conduct a structured training program for GWSSB's nominated departmental staff throughout the O&M period, consistent with the capacity-building objectives set out in Part H.3 of the BASE_DOC, covering plant operation, water quality monitoring, SCADA system use, and emergency response procedures.

The Defect Liability Periods of 24 months (civil) and 18 months (mechanical/electrical) from Provisional Acceptance, as required under SCC-7, are accepted, running concurrently with the commencement of the O&M period. Equipment-specific manufacturer warranties, where they extend beyond the contractual DLP, will be passed through to GWSSB along with the relevant OEM warranty certificates.

At the end of the 10-year O&M period, HydroTech Engineering Pvt. Ltd. commits to handing back the Works in the condition specified in SCC-14, with a joint condition assessment to be conducted in the final year of the O&M period to identify and complete any residual-life restoration works required.

24. Deviations from Tender Requirements

Clause / Requirement	Deviation Stated	Evaluation Committee Assessment
Disinfection System (Part E.5 / D.7)	Sodium hypochlorite dosing proposed in place of gas chlorination; not clearly flagged as a deviation in the Bid Submission Form as required by ITB-10	Major - technically permissible per D.7 if properly declared and justified, but the failure to explicitly declare it is itself a procedural non-compliance requiring Evaluation Committee determination
Filter Media Depth (Part E.4)	Sand depth 55cm and anthracite depth 25cm offered, both below the specified minimum ranges	Major - directly affects filtered water quality and filter run length; not acceptable without redesign or explicit, justified deviation with performance guarantee
Execution Schedule	Compressed schedule of 552 days (92% of 600-day base) offered	Positive deviation (favorable) - subject to verification of resource loading feasibility given the reduced manpower shown in Section 13

25. Clarifications Sought & Bidder Responses

Ref.	Clarification Sought by GWSSB	Bidder Response
CL-1	Clarify basis for offering sodium hypochlorite dosing and confirm whether this should be treated as a formal deviation	Bidder confirmed this was an inadvertent omission in the Bid Submission Form and requested it be treated as a formally declared deviation; cost and O&M implications (chemical cost, storage) to be re-evaluated
CL-2	Justify filter media depths below Part E.4 minimum specification	Bidder proposed a higher filtration rate design (7.5 m ³ /m ² /hr) claiming equivalent performance with reduced media depth; GWSSB Process Cell to independently verify against CPHEEO criteria (Annexure L-7)
CL-3	Confirm feasibility of compressed schedule given reduced manpower deployment shown in Section 13	Bidder confirmed additional 20% surge manpower would be mobilized during peak erection phase beyond the base plan shown

26. Priced Bill of Quantities (BOQ) - Stage 1 (Initial D&B Price, submitted 05-Mar-2026)

Description	Unit	Qty	Rate (₹)	Amount (₹)
Site clearance, grading and levelling	sqm	45000	₹81	₹36,45,000
Excavation for structures (ordinary soil)	cum	32000	₹198	₹63,36,000
Excavation for rising main trench	cum	68000	₹225	₹1,53,00,000
Dewatering during excavation	LS	1	₹76,50,000	₹76,50,000
PCC M-10 for foundation bed	cum	850	₹4,770	₹40,54,500
RCC M-25 for water-retaining structures	cum	3600	₹8,280	₹2,98,08,000
RCC M-20 for non-water-retaining structures	cum	2100	₹7,020	₹1,47,42,000
Reinforcement steel (Fe-500D)	MT	1850	₹61,200	₹11,32,20,000
Crystalline/membrane waterproofing	sqm	18000	₹765	₹1,37,70,000
Structural steel (galvanized)	MT	320	₹1,18,800	₹3,80,16,000
Masonry works	cum	2400	₹4,320	₹1,03,68,000
Plastering (incl. acid-resistant)	sqm	28000	₹288	₹80,64,000
Flooring (non-slip/chemical-resistant)	sqm	12000	₹855	₹1,02,60,000
Roofing	sqm	9500	₹1,485	₹1,41,07,500
Internal roads (bituminous)	sqm	18000	₹1,305	₹2,34,90,000
Storm water drainage	rmt	2800	₹1,665	₹46,62,000
Boundary wall	rmt	1600	₹3,780	₹60,48,000
Landscaping	sqm	15000	₹342	₹51,30,000
Chlorine safety structure	LS	1	₹1,08,00,000	₹1,08,00,000
Raw Water Pumping Station - civil	LS	1	₹7,65,00,000	₹7,65,00,000
Raw water pumps (incl. motor)	No.	6	₹61,20,000	₹3,67,20,000
Raw water rising main pipeline	rmt	2800	₹11,250	₹3,15,00,000
Screening equipment	No.	3	₹19,80,000	₹59,40,000
Clarifloculator mechanism	No.	4	₹1,66,50,000	₹6,66,00,000
Tube settler modules	sqm	3200	₹3,780	₹1,20,96,000
RGF civil works (16 beds)	LS	900	₹13,05,00,000	₹1,17,45,00,00,000
Filter media (sand + anthracite)	cum	1200	₹7,650	₹91,80,000
Filter underdrain system	sqm	720	₹11,250	₹81,00,000
Backwash pumps & blowers	Set	3	₹28,80,000	₹86,40,000
Chlorination system (complete)	LS	1	₹3,78,00,000	₹3,78,00,000
Gas leak detection & safety system	Set	1	₹76,50,000	₹76,50,000
Sludge thickener	No.	2	₹1,08,00,000	₹2,16,00,000
Sludge dewatering system	No.	1	₹3,42,00,000	₹3,42,00,000
Sludge drying beds	sqm	4500	₹1,980	₹89,10,000
Clear Water Reservoir - civil	LS	1	₹14,85,00,000	₹14,85,00,000
Clear Water Pumping Station - civil	LS	1	₹6,48,00,000	₹6,48,00,000
Clear water pumps (incl. motor)	No.	7	₹46,80,000	₹3,27,60,000
Clear water rising main (8.5km)	rmt	8500	₹34,650	₹29,45,25,000
Surge protection system	Set	1	₹1,66,50,000	₹1,66,50,000
HT substation	LS	1	₹2,88,00,000	₹2,88,00,000
LT MCC panels (all areas)	Set	12	₹14,85,000	₹1,78,20,000
DG sets (2x100%)	No.	2	₹76,50,000	₹1,53,00,000
Power & control cabling	rmt	24000	₹378	₹90,72,000
Earthing & lightning protection	LS	1	₹85,50,000	₹85,50,000
SCADA/PLC system	LS	1	₹2,56,50,000	₹2,56,50,000
Field instrumentation	Set	1	₹1,66,50,000	₹1,66,50,000
Fiber optic backbone	rmt	3200	₹765	₹24,48,000
Laboratory equipment (complete)	LS	1	₹1,12,50,000	₹1,12,50,000
Administrative building	sqm	850	₹28,800	₹2,44,80,000
Security cabin & CCTV	LS	1	₹37,80,000	₹37,80,000
Fire fighting system	LS	1	₹76,50,000	₹76,50,000

Description	Unit	Qty	Rate (₹)	Amount (₹)
HVAC system	LS	1	₹55,80,000	₹55,80,000
Type & routine testing charges	LS	1	₹1,08,00,000	₹1,08,00,000
Third-party inspection charges	LS	1	₹76,50,000	₹76,50,000
Commissioning & PGT support	LS	1	₹1,35,00,000	₹1,35,00,000
As-built documentation & O&M manuals	LS	1	₹37,80,000	₹37,80,000
Training of Employer's personnel	LS	1	₹28,80,000	₹28,80,000
Mandatory spares (2-year)	LS	1	₹3,42,00,000	₹3,42,00,000
Insurance (CAR+TPL, D&B phase)	LS	1	₹1,66,50,000	₹1,66,50,000
Mobilization & demobilization	LS	1	₹1,12,50,000	₹1,12,50,000

Total D&B Base Price (Stage 1): ₹1,18,98,98,82,000 | GST @ 18%: ₹21,41,81,78,760 | Total D&B Bid Value (Stage 1): ₹1,40,40,80,60,760

27. Priced Bill of Quantities (BOQ) - Stage 2 (Final D&B Price)

No base price revision offered. Bidder submitted an alternate-conforming price addendum reflecting gas chlorination (in place of sodium hypochlorite) and filter media depths within the Part E.4 specified range, to be considered only if the Evaluation Committee does not accept the deviations in Sections 18; this addendum shows a price increase of 7.8% over the originally quoted base price.

Description	Unit	Qty	Rate (₹)	Amount (₹)
Site clearance, grading and levelling	sqm	45000	₹81	₹36,45,000
Excavation for structures (ordinary soil)	cum	32000	₹198	₹63,36,000
Excavation for rising main trench	cum	68000	₹225	₹1,53,00,000
Dewatering during excavation	LS	1	₹76,50,000	₹76,50,000
PCC M-10 for foundation bed	cum	850	₹4,770	₹40,54,500
RCC M-25 for water-retaining structures	cum	3600	₹8,280	₹2,98,08,000
RCC M-20 for non-water-retaining structures	cum	2100	₹7,020	₹1,47,42,000
Reinforcement steel (Fe-500D)	MT	1850	₹61,200	₹11,32,20,000
Crystalline/membrane waterproofing	sqm	18000	₹765	₹1,37,70,000
Structural steel (galvanized)	MT	320	₹1,18,800	₹3,80,16,000
Masonry works	cum	2400	₹4,320	₹1,03,68,000
Plastering (incl. acid-resistant)	sqm	28000	₹288	₹80,64,000
Flooring (non-slip/chemical-resistant)	sqm	12000	₹855	₹1,02,60,000
Roofing	sqm	9500	₹1,485	₹1,41,07,500
Internal roads (bituminous)	sqm	18000	₹1,305	₹2,34,90,000
Storm water drainage	rmt	2800	₹1,665	₹46,62,000
Boundary wall	rmt	1600	₹3,780	₹60,48,000
Landscaping	sqm	15000	₹342	₹51,30,000
Chlorine safety structure	LS	1	₹1,08,00,000	₹1,08,00,000
Raw Water Pumping Station - civil	LS	1	₹7,65,00,000	₹7,65,00,000
Raw water pumps (incl. motor)	No.	6	₹61,20,000	₹3,67,20,000
Raw water rising main pipeline	rmt	2800	₹11,250	₹3,15,00,000
Screening equipment	No.	3	₹19,80,000	₹59,40,000
Clariflocculator mechanism	No.	4	₹1,66,50,000	₹6,66,00,000
Tube settler modules	sqm	3200	₹3,780	₹1,20,96,000
RGF civil works (16 beds)	LS	900	₹13,05,00,000	₹1,17,45,00,00,000
Filter media (sand + anthracite)	cum	1200	₹7,650	₹91,80,000
Filter underdrain system	sqm	720	₹11,250	₹81,00,000
Backwash pumps & blowers	Set	3	₹28,80,000	₹86,40,000
Chlorination system (complete)	LS	1	₹3,78,00,000	₹3,78,00,000
Gas leak detection & safety system	Set	1	₹76,50,000	₹76,50,000
Sludge thickener	No.	2	₹1,08,00,000	₹2,16,00,000
Sludge dewatering system	No.	1	₹3,42,00,000	₹3,42,00,000
Sludge drying beds	sqm	4500	₹1,980	₹89,10,000
Clear Water Reservoir - civil	LS	1	₹14,85,00,000	₹14,85,00,000
Clear Water Pumping Station - civil	LS	1	₹6,48,00,000	₹6,48,00,000
Clear water pumps (incl. motor)	No.	7	₹46,80,000	₹3,27,60,000
Clear water rising main (8.5km)	rmt	8500	₹34,650	₹29,45,25,000
Surge protection system	Set	1	₹1,66,50,000	₹1,66,50,000
HT substation	LS	1	₹2,88,00,000	₹2,88,00,000
LT MCC panels (all areas)	Set	12	₹14,85,000	₹1,78,20,000
DG sets (2x100%)	No.	2	₹76,50,000	₹1,53,00,000
Power & control cabling	rmt	24000	₹378	₹90,72,000
Earthing & lightning protection	LS	1	₹85,50,000	₹85,50,000
SCADA/PLC system	LS	1	₹2,56,50,000	₹2,56,50,000
Field instrumentation	Set	1	₹1,66,50,000	₹1,66,50,000

Description	Unit	Qty	Rate (₹)	Amount (₹)
Fiber optic backbone	rmt	3200	₹765	₹24,48,000
Laboratory equipment (complete)	LS	1	₹1,12,50,000	₹1,12,50,000
Administrative building	sqm	850	₹28,800	₹2,44,80,000
Security cabin & CCTV	LS	1	₹37,80,000	₹37,80,000
Fire fighting system	LS	1	₹76,50,000	₹76,50,000
HVAC system	LS	1	₹55,80,000	₹55,80,000
Type & routine testing charges	LS	1	₹1,08,00,000	₹1,08,00,000
Third-party inspection charges	LS	1	₹76,50,000	₹76,50,000
Commissioning & PGT support	LS	1	₹1,35,00,000	₹1,35,00,000
As-built documentation & O&M manuals	LS	1	₹37,80,000	₹37,80,000
Training of Employer's personnel	LS	1	₹28,80,000	₹28,80,000
Mandatory spares (2-year)	LS	1	₹3,42,00,000	₹3,42,00,000
Insurance (CAR+TPL, D&B phase)	LS	1	₹1,66,50,000	₹1,66,50,000
Mobilization & demobilization	LS	1	₹1,12,50,000	₹1,12,50,000

Final D&B Base Price (Stage 2): ₹1,18,98,98,82,000 | GST @ 18%: ₹21,41,81,78,760 | Final D&B Bid Value (Stage 2): ₹1,40,40,80,60,760

28. O&M Price Schedule (10-Year, Year-wise) & Combined Cost Evaluation

The following table sets out the bidder's quoted O&M fee for each of the 10 years of the O&M period, escalated per the bidder's proposed basis (subject to the WPI-linked cap of 7% p.a. from Year 2, per SCC-9), and the resulting Net Present Value (NPV) at the 8% discount rate specified in Part G of the BASE_DOC.

O&M Year	Annual Fee (₹)
Year 1	₹7,10,00,000
Year 2	₹7,56,15,000
Year 3	₹8,05,29,975
Year 4	₹8,57,64,423
Year 5	₹9,13,39,111
Year 6	₹9,72,76,153
Year 7	₹10,35,99,103
Year 8	₹11,03,33,045
Year 9	₹11,75,04,693
Year 10	₹12,51,42,498

Total 10-Year O&M Fee (NPV @ 8%): ₹66,72,29,230

**COMBINED EVALUATED COST (Final D&B Bid Value + NPV of 10-Year O&M Fee):
₹1,41,07,52,89,990**

29. Payment Terms Compliance Statement

HydroTech Engineering Pvt. Ltd. confirms acceptance of the Milestone Payment Schedule set out in Part J of the BASE_DOC for the D&B phase, and the O&M fee payment mechanism (quarterly in arrears, subject to performance guarantee compliance per Part H) for the O&M phase, with payments released against certification by GWSSB's Engineer-in-Charge.

29A. Sample Treated Water Quality Test Report Format (IS 10500:2012 Parameters)

For the Employer's reference, the following illustrates the parameter set and reporting format proposed for the monthly third-party treated water quality test report to be submitted throughout the O&M period, in compliance with IS 10500:2012.

Parameter	IS 10500:2012 Acceptable Limit	Test Method Reference
Turbidity (NTU)	1 (5 if no alternate source)	IS 3025 (Part 10)
pH	6.5 - 8.5	IS 3025 (Part 11)
Total Dissolved Solids (mg/l)	500 (2000 max)	IS 3025 (Part 16)
Total Hardness as CaCO ₃ (mg/l)	200 (600 max)	IS 3025 (Part 21)
Chlorides (mg/l)	250 (1000 max)	IS 3025 (Part 32)
Sulphate (mg/l)	200 (400 max)	IS 3025 (Part 24)
Fluoride (mg/l)	1.0 (1.5 max)	IS 3025 (Part 60)
Nitrate (mg/l)	45	IS 3025 (Part 34)
Iron (mg/l)	0.3 (1.0 max)	IS 3025 (Part 53)
Residual Free Chlorine (mg/l)	0.2 minimum	IS 3025 (Part 26)
Total Coliform (MPN/100ml)	Shall not be detectable in any 100ml sample	IS 1622
E. coli (MPN/100ml)	Shall not be detectable in any 100ml sample	IS 1622
Arsenic (mg/l)	0.01 (0.05 max)	IS 3025 (Part 37)
Total Alkalinity (mg/l)	200 (600 max)	IS 3025 (Part 23)
Copper (mg/l)	0.05 (1.5 max)	IS 3025 (Part 42)
Zinc (mg/l)	5	IS 3025 (Part 49)
Manganese (mg/l)	0.1 (0.3 max)	IS 3025 (Part 59)
Aluminium (mg/l)	0.03 (0.2 max)	IS 3025 (Part 55)
Cyanide (mg/l)	0.05	IS 3025 (Part 27)
Total Chromium (mg/l)	0.05	IS 3025 (Part 52)

30. List of Annexures

- Annexure-A: Bid Submission Form (signed)
- Annexure-B: EMD Instrument (Bank Guarantee)
- Annexure-C: Project Completion Certificates, O&M Performance Records & Client Reference Letters
- Annexure-D: Process Design Calculations (preliminary)
- Annexure-E: OEM Authorization Letters & Approved Makes Cross-Reference
- Annexure-F: Audited Financial Statements (FY 2022-23 to 2024-25) & CA Certificate
- Annexure-G: Level-3 Project Schedule (Bar Chart)
- Annexure-H: ISO 9001/14001/45001 Certificates
- Annexure-I: Statutory Registrations (PAN, GST, EPFO, PCB Consents)
- Annexure-J: Power of Attorney
- Annexure-K: Blacklisting Declaration (Notarized)
- Annexure-L: General Arrangement (GA) Drawing - Preliminary
- Annexure-M: Process Flow Diagram (PFD) - Preliminary
- Annexure-N: Equipment Datasheets (OEM catalogues)
- Annexure-O: Integrity Pact (signed)
- Annexure-P: O&M Staffing Plan - Detailed

31. Compliance Undertaking

We, HydroTech Engineering Pvt. Ltd., hereby undertake that the information furnished in this bid is true and correct to the best of our knowledge, and that we accept the Instructions to Bidders, General Conditions, Special Conditions and all other terms of Tender GWSSB/WTP-SURAT/2026/DBO-009 in full, except for the deviations explicitly listed in Section 24 above. We understand that any misrepresentation discovered at any stage may result in disqualification of this bid and forfeiture of EMD/Performance Security, as applicable.

32. Evaluation Committee Remarks

Lowest quoted Combined Cost among all bidders but carries two major, potentially disqualifying technical deviations (undeclared disinfection system change, under-spec filter media depth) that require Evaluation Committee adjudication. O&M experience just meets the 3-year minimum with no margin. If deviations are not waived, the bidder's own conforming-alternate price (7.8% higher) would apply for comparison.